

MSDS 535 – Week 14

Sections 12.3 and 12.4

Overview and Objectives

- Data mining methodologies and systems.
- Data mining, people, and society.

Structuring Unstructured Data

- Conversion of unstructured data into structured information is vital for effective knowledge extraction.
- Leveraging existing knowledge-bases like Wikipedia and pre-trained models is fundamental.
- The approach involves weakly supervised methods for taxonomy creation and expansion.

Structuring Unstructured Data

- Taxonomies must be adaptable, aligning with specific applications and datasets.
- Generating, refining, and expanding taxonomies are essential tasks in knowledge extraction.

Weakly Supervised Text Processing

- Weakly supervised text classification methods rely on limited labeled data and extensive unlabeled datasets.
- Hierarchical multilabel text classification, based on taxonomy structures, is a developed approach.
- Classification considers numerous classes and multiple labels, often organized within a taxonomy.

Weakly Supervised Text Processing

- These methods employ embedding techniques and pre-trained language models for classification.
- Cost-effectiveness drives the preference for weakly supervised over fully supervised classification.

Fine-Grained Entity Recognition

- Fine-grained entity recognition identifies specific roles entities play in different contexts.
- Distant supervision is used to determine the context-appropriate types for entities.
- Methods like ClusType utilize clustering and relation phrase propagation for entity context identification.

Fine-Grained Entity Recognition

- Taxonomy-guided supervision enhances text classification and named entity recognition.
- The context-sensitive entity recognition enriches information extraction from text.

Knowledge Graph and Information Network Construction

- Construction of knowledge graphs involves entities, attributes, and edges, often with taxonomy integration.
- Distant or weak supervision aids in constructing these graphs from extensive text data.
- Utilizing local knowledge graphs under specific conditions proves more useful than a single global graph.

Knowledge Graph and Information Network Construction

- Taxonomy-guided text classification assists in allocating information to corresponding nodes or subgraphs.
- Different associations of entities from massive text are structured within local knowledge graphs.

Contributions and Studies

- Recent studies focus on weakly supervised approaches for various tasks in knowledge extraction.
- Methods like AutoPhrase and UCPhrase focus on unsupervised phrase mining for entity extraction.
- Several studies concentrate on ontology-guided fine-grained named entity recognition.

Contributions and Studies

- Embedding-based taxonomy construction and expansion methods are being extensively explored.
- Overall, research continually advances weakly supervised approaches for structured knowledge extraction.

The Importance of Data Augmentation

- Limited labeled data poses challenges in building high-performing data mining models.
- Various factors like data privacy, cost, imbalance, and continuous data growth restrict labeled data availability.
- Data augmentation enriches training data, aiding models in extracting more meaningful features.

The Importance of Data Augmentation

- The concept relies on invariance properties observed in different types of data.
- Augmentation methods are categorized into basic and learning-based techniques.

Basic Data Augmentation Methods

- Basic methods utilize human recognition systems to generate verified augmented samples.
- Operations like flipping, rotation, and color space adjustment enhance data robustness.
- These methods are applicable across various data types such as images and audio.

Basic Data Augmentation Methods

- The goal is to ensure data samples retain their labels after augmentation.
- Basic augmentation techniques amplify the robustness of models akin to human recognition systems.

Learning-Based Augmentation Approaches

- Augmentation policy learning automates the search for the best augmentation strategies.
- Methods like AutoAugment use sequences of operations and magnitudes to define policies.
- GANs-based methods leverage adversarial networks to generate versatile data samples.

Learning-Based Augmentation Approaches

- GANs learn from abundant samples to enrich classes with limited data, addressing data scarcity.
- Adversarial training introduces imperceptible perturbations to enhance model robustness.

Contrasting Augmentation Methods

- Basic methods rely on predefined operations while learning-based ones employ parameterized components.
- Augmentation policy learning may face efficiency challenges due to policy model updates.
- GANs-based methods instantiate transfer learning by constructing data manifolds from abundant to limited samples.
- Adversarial training focuses on improving model robustness against adversarial attacks.

Bibliographic Insights

- Various methods like AutoAugment and DAGAN automate augmentation and GANs incorporation.
- Augmentation benefits diverse fields like medical imaging and machine fault diagnosis.
- GANs-based approaches address challenges in scenarios with limited data.
- Augmentation remains underexplored in graph-structured data despite solutions like GraphSGAN.

Transitioning from Correlation to Causality

- Data mining primarily involves correlation analysis rather than causality exploration.
- Correlation studies statistical associations between observed variables; causality focuses on causal relations.
- The distinction is illustrated with an example concerning temperature, electric bills, and food expenses.

Challenges and Tasks in Causality Analysis

- Causality analysis faces challenges in discovery and inference, differing from correlation studies.
- Discovery involves qualitative analysis: determining which variables to manipulate.
- Inference is quantitative: estimating changes when modifying variables, often requiring complex experiments.

Applications and Importance of Causality Analysis

- Enhanced interpretability of deep learning models through causality discovery.
- Mitigation of biased outcomes related to sensitive variables in systems like university admissions.
- Improved model robustness and avoidance of unfair results are among the benefits.

Taxonomy of Causality Analysis Methods

- Traditional methods include constraint-based algorithms, score-based algorithms, and functional causal models.
- Big data challenges are addressed by restricting the search space for high-dimensional situations.
- Causality inference involves regression adjustment, propensity score methods, and covariate balancing methods.
- Methods like instrumental variable (IV) and front-door criterion handle cases with unobserved confounders.

Trends and Ongoing Challenges

- Current trends aim to replace traditional methods' prior knowledge with data-driven strategies.
- Integration of deep learning methods into causal models to enhance performance is becoming prevalent.
- Complex treatment scenarios pose challenges, remaining areas open for exploration in causality analysis.

Bibliographic Insights

- Various methodologies such as structural equation models and Gaussian noise incorporation facilitate causal graph generation.
- Advanced learning methods like neural networks and ensemble models enhance causality inference.
- Many challenges in diverse applications of causality analysis, including interpretation of black-box models, model robustness, and fairness, await exploration.

Networks: A Contextual Framework

- Networks (graphs) are pivotal in data mining, connecting various data types and algorithms.
- "Network-of-X" concept links entities, datasets, or models via an underlying contextual network.

Diverse Network Contexts

- Networks of Time Series involve interconnected devices monitoring various signals in surveillance systems, aiding in detecting abnormalities by leveraging contextual correlations.
- Network of Networks comprises domain-specific collaboration networks connected by a research domain network, enhancing mining tasks across domains through hierarchical structures and cross-network consistency.

Diverse Network Contexts

- Networks of Data Mining Models link regression models through collaboration networks, improving prediction performance by modeling collaboration among task-specific teams.
- Devices in a sensor network capture signals, and the underlying contextual network aids in detecting abnormalities by mining multiple intercorrelated signals together.

Diverse Network Contexts

- Collaboration networks in research domains are connected by a domain network, allowing for multiresolution characteristics and cross-network consistency in complex datasets.
- Regression models on task-specific teams are linked through collaboration networks, enhancing prediction performance by leveraging the contextual network.

Unexplored Territory and Future Directions

- Contextual network quality impacts overall performance; optimal similarity measures are pivotal.
- Expanding "X" options for broader applications (knowledge graphs, heterogeneous networks).
- Integration of other data mining problems (adversarial learning, fairness) with Network-of-X.

Bibliographic Insights

- Methods like dynamic contextual matrix factorization and sequence-to-sequence architectures enhance time-series analysis.
- Cross-network consistency principles and clustering methods aid in domain-specific network analysis.
- Encouraging consistency between closely related predictive models for improved performance.

Auto-ML Introduction

- In various fields, data mining and machine learning models face diverse settings and demand specialized models due to different tasks and data complexities.
- Domain experts often lack detailed knowledge about building and fine-tuning models for specific tasks, leading to the need for automated solutions.

Auto-ML Categories

- Automated Feature Engineering (Auto-FE) focuses on detecting and selecting essential features automatically.
- Automated Model and Hyperparameter Tuning (Auto-MHT) involves the automatic creation of models and fine-tuning of hyperparameters.

Techniques in Auto-ML

- Reinforcement Learning (RL) is utilized to construct a transformation graph for optimizing feature performance.
- Evolutionary Algorithms (EA) create a tree structure to facilitate feature transformation.
- Bayesian Optimization (BO) leverages probabilistic models to find the best hyperparameter settings.

Techniques in Auto-ML

- RL and Genetic Algorithms (GA) are employed in hyperparameter tuning and model design.
- EA concentrates on finding optimal model architectures, while GA encodes these architectures into a continuous space.

Auto-ML Challenges

- The absence of standardized benchmarks for Auto-FE and Auto-MHT poses a significant challenge.
- Deep learning Auto-MHT methods often consume substantial time and resources, affecting efficiency.
- Incorporating human knowledge effectively into the Auto-ML process remains a challenge.

Auto-ML Challenges

- Interpreting the rationale behind chosen features and architectures poses a challenge for user understanding.
- Limited availability of Auto-ML frameworks catering to complex tasks beyond classification and regression.

Bibliographic Notes

- Auto-FE methods utilize RL, EA for automated data transformation processes.
- Auto-MHT techniques employ BO, RL, EA for automatic model and hyperparameter optimization.
- Diverse methods focus on finding optimal neural architectures for deep learning models.

Privacy Concerns in Data Mining

- Privacy risks during data analysis can have severe consequences for individuals, service providers, and society.
- Misuse or breach of data can lead to re-identification, lawsuits, and ethical debates on using obtained knowledge.
- Addressing these concerns becomes increasingly critical with the growing impact of data mining on individuals' privacy.

Anonymization Methods: k-Anonymity

- k-Anonymity groups data to protect identifying attributes but remains vulnerable to background knowledge attacks.
- Challenges arise in balancing data utility and privacy preservation while using k-anonymity techniques effectively.

Differential Privacy as Privacy Protection

- Differential privacy ensures aggregate analysis while obscuring individual contributions, preventing re-identification.
- Parameters like epsilon (ϵ) in differential privacy algorithms govern the balance between utility and privacy, requiring careful consideration.
- Differential privacy offers a robust privacy-preserving approach but faces complexity in applying it universally.

Challenges in Implementing Differential Privacy

- Calculating global sensitivity for some functions remains complex, affecting the implementation of differential privacy.
- Finding an equilibrium between privacy preservation and data utility is an ongoing challenge in deploying these techniques effectively.
- The tradeoff between privacy and utility is a central concern and requires continual research and refinement.

Federated Learning for Privacy Preservation

- Federated learning enables collaborative data mining while preserving individual data privacy through local computations.
- Balancing data aggregation for analysis without revealing sensitive information remains a key focus in federated learning.
- Methods like federated analytics provide promising avenues for collaboration without exposing detailed data records.

Legal and Societal Efforts for Privacy

- Legislations such as GDPR in the EU and HIPAA in the US aim to safeguard data privacy and encourage anonymization.
- Privacy preservation in data mining transcends technical aspects, necessitating legal and societal measures to protect individual privacy rights.
- Efforts to regulate and enforce privacy policies underscore the importance of ethical considerations in data mining practices.

Enhancing Algorithms with Human Collaboration

- Human-algorithm interaction seeks to optimize performance by blending human intelligence and data mining algorithms.
- Perspectives include leveraging human intellect for challenging tasks, improving algorithms through human intervention, refining query strategies, and fostering effective teamwork.

Crowdsourcing

- Crowdsourcing taps into human intelligence for tasks beyond machine capabilities.
- Applications range from image recognition to product ratings and summarization.
- Generates high-quality labeled data samples and evaluates learning models or interprets predictions in critical domains like medical diagnosis.
- Aids in debugging AI systems and evaluating the interpretability of model predictions.

Human-in-the-Loop (HITL)

- Integrates human intelligence into learning algorithm development.
- Aids model interpretability, provides evaluations for explanation quality, and annotates complex data like medical images.
- Expertise in medical science plays a crucial role in annotating medical data for accurate diagnosis.

Machine-in-the-Loop

- Focuses on using learning algorithms to assist humans in tasks like creative writing and drawing.
- Reduces the need for labeled data in active learning scenarios, leveraging AI for creative idea generation.

Human-Machine Teaming (HMT)

- Emphasizes collaboration between humans and algorithms for superior performance.
- Challenges include modeling coordination, predicting team performance, and optimizing human-machine interactions for effective results.

Bibliographic Insights

- Various studies explore crowdsourcing applications, from image annotation to improving medical diagnosis reliability.
- Incorporating human intelligence enhances clustering algorithms, aids in machine teaching, and supports explainability in machine learning for better decision-making.
- Human input is critical for tasks requiring domain expertise, such as medical data annotation and model interpretability in crucial domains.

Ubiquitous Decision Making

- Data mining enables automated decision-making in diverse domains like social media, recruitment, and courts, offering recommendations and predictions based on user preferences and social networks.

Algorithmic Fairness

- Fairness can be defined in various ways, including group, individual, and counterfactual fairness.
- Strategies for fair learning include preprocessing, optimization during training, and postprocessing to mitigate bias.
- Active research explores biases in data generation and long-term effects of feedback loops on fairness.

Interpretability

- Challenges the opacity of data mining models, highlighting the need for understanding and transparency.
- Concerns arise regarding trust, reliability, and contestability when models lack human interpretability.
- Various techniques aim to enhance interpretability, from visualizing feature statistics to identifying key data points for a comprehensible model.

Robustness

- Focuses on model stability amidst unintentional noise or adversarial attacks.
- Illustrates the vulnerability of models like Convolutional Neural Networks to imperceptible modifications.
- Adversarial attack taxonomy outlines methods to challenge models, prompting reactive and proactive defense mechanisms.

Bibliographic Insights

- Research endeavors tackle discrimination mitigation, feature obfuscation, and ensuring fairness in model predictions.
- Extensive studies explore interpretable machine learning methods, emphasizing explainable decision-making models.
- Strategies for defending against adversarial attacks range from detection and reconstruction to robustifying classifiers.

Lead Water Pipe Detection

- Flint, Michigan's water contamination had severe health consequences.
- The ActiveRemediation project, led by the University of Michigan, uses XGBoost and Bayesian spatial models to replace hazardous pipes with 70% accuracy.

Diverse Societal Applications

- Data mining aids education, public health, combatting misinformation, social care, urban planning, safety, and transportation.
- Models like Random Forests identify at-risk students, EpiDeep predicts epidemics, and SVMs detect rumors.
- Techniques encompass Graph Attention Networks and Transfer Learning.

Research Challenges

- Scarcity of data limits supervised methods, prompting use of unsupervised or transfer learning.
- Standard evaluation metrics may fall short; domain-specific measures are crucial.
- Involving domain experts is crucial; effective interaction between humans and algorithms is vital.
- Sustainable deployment faces hurdles due to collaboration willingness and funding status.

Bibliographic Insights

- Studies range from detecting hazardous pipes to predicting epidemics.
- Techniques include LSTM for irregular patient records, SVMs for rumor detection, and FLORAL for data insufficiency in urban computing.
- Reinforcement learning frameworks effectively coordinate ride-sharing dynamics.